

Robots refuel satellites!

NASA is making a robotic spaceship to refuel satellites, that might make refuelling much easier.
<http://dgit.in/28SCsXJ>



Pluto has an ocean?

Pluto might still have an liquid or semi liquid ocean under it's icy crust.
<http://dgit.in/294xPu3>

From the labs

to be yet. These are extremely quick at proliferating but are unwieldy to use because it is difficult to engineer these cells to take the structure we want. Differentiated muscle fibres have the exact texture and flavour we need but do not proliferate at all and can't, as such, be grown. A good compromise is to pick cells that are somewhere in between.

- **A growth medium:** The medium must be able to support cell growth by providing the cells with all the nutrients it required directly. Of course, ideally, you would require this medium to not be derived directly from an animal.
- **Scaffold:** This is potentially the difficult part. The scaffold is a surface that the cells can attach to. Apart from needing it to be edible so that the scaffold can become a part of the final product, it is also important that the scaffold have the same kind of texture that we require our end product to have. It is also a good idea to have a scaffold that can move in response to some kind of stimulus so that the muscle fibres attached to the scaffold can stretch and elongate as required. The scaffold along with the cells attached to it give the 3D structure that goes into the final product.
- **Bioreactor:** This is where all of the ingredients are put together and grown. There are many specific requirements that the reactor must satisfy, a high school petri dish will not suffice.

There has been different levels of success and progress in each of these categories but so far there has been no major scientific breakthroughs but scientists are, for the first time, aware of what exactly is necessary to make artificial meat a reality.



Artificially produced meat

The Dutch burger

€250,000 can buy a pretty posh house, a really fast car, or if you're Google executive Sergey Brin and the person you're giving the money to is a Dutch university, it can buy you a hamburger. Sergey Brin was pushed to sponsor this project by what he felt were erroneous ideas that people had about the meat industry. Encouraging people to not eat meat is not a viable option as the tendency to eat meat is 'ingrained deep in our evolution', according to Prof Wrangham, primatologist from Harvard University. With such severe constraints, growing artificial meat is a pressing need of our time and Brin is one of the first people to recognise this.

The most expensive hamburger in the world was a three month long project headed by Dr Mark Post at Maastricht University that began with the extraction of stem cells from two cows, a blanc-blue belge and a blond acquitaine. Dr Mark Post and his team at Maastricht University used these cells to grow 20,000 muscle fibres in individual culture wells, each one a tiny hoop of greyish-white protein suspended in a gel-like growth medium that contained antibiotics and a serum extracted from cow foetuses. The cells were allowed to grow for a few weeks to gain some size. After this, each fibre was removed individually, cut open and straightened out in a long string. These fibres form the base of the final patty. Compressed together, these fibres are then coloured with beetroot juice to give it the colour of ground beef, mixed with saffron, breadcrumbs and a few other ingredients to bind it together. "It's really just proof of concept right now, we're trying to create the first cultured beef hamburger," said Brin in a film to mark the tasting event in London on Monday. "From there I'm optimistic that we can really scale by leaps and bounds."

This burger however, is pure protein and has no fats, a fact that has not escaped the notice of tasters. "The texture, the mouthfeel has a feel like meat," journalist Schonwald put it. "The absence is ... fat. It's a leanness. But the bite feels like a conventional hamburger. It's kind of an unnatural experience" The next test therefore, for Post and his team, is to incor-

porate fat and to reduce costs. There has been huge progress in the past couple of years with the team able to reduce to cost to \$12 per patty. This is still quite expensive but at least the price is on the same scale as conventional burgers.

Other lab grown vegan wonders

Substitutes are being created for virtually all animal based products. One of the most interesting projects in this endeavour is titled Real Vegan Cheese, a group of biohackers that are looking to create milk, and consequently all dairy based products such as cheese and butter, directly from DNA of cow's milk protein. One day, they hope to achieve the proficiency to be able to make complicated cheeses from one DNA extract.



The final burger. Not quite a whopper yet

The lab-grown-food sector is growing quickly. Along with Brin, other big-name backers include Microsoft co-founder Bill Gates and PayPal co-founder Peter Thiel.

"The global food supply chain is a multitrillion-dollar industry," says Arvind Gupta, founder of IndieBio, an accelerator that exclusively funds synthetic-biology startups. "That's the market we're thinking about disrupting."

The future

There is a lot of scope and funding for lab grown animal product substitutes. Despite a few criticisms, notably from activists who worry about the impact that this may have on livestock farmers, the industry is one that is set to witness breakthrough progress in the next few years. The day may not be far when a Maharaja Mac is grown in a lab and we'd be none the wiser. 